

Binomial Coefficient Identities

1. Fibonacci Binomial Identity

$$\sum_{0 \leq k \leq n} \binom{n-k}{k} = \text{Fib}_{n+1}$$

2. Symmetry Rule

$$\binom{n}{k} = \binom{n}{n-k}$$

3. Pascal's Recurrence

$$\binom{n}{k} + \binom{n}{k+1} = \binom{n+1}{k+1}$$

4. Absorption Identity

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

5. Factoring In

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}$$

6. Sum of Binomial Coefficients

$$\sum_{i=0}^n \binom{n}{i} = 2^n$$

7. Sum of Even Binomial Coefficients

$$\sum_{i \geq 0} \binom{n}{2i} = 2^{n-1}$$

8. Sum of Odd Binomial Coefficients

$$\sum_{i \geq 0} \binom{n}{2i+1} = 2^{n-1}$$

9. Alternating Sum Identity

$$\sum_{i=0}^k (-1)^i \binom{n}{i} = (-1)^k \binom{n-1}{k}$$

10. Hockey-Stick Identity

$$\sum_{i=0}^k \binom{n+i}{i} = \sum_{i=0}^k \binom{n+i}{n} = \binom{n+k+1}{k}$$

11. Weighted Linear Sum

$$\sum_{i=1}^n i \binom{n}{i} = n 2^{n-1}$$

12. **Weighted Quadratic Sum**

$$\sum_{i=1}^n i^2 \binom{n}{i} = (n + n^2)2^{n-2}$$

13. **Vandermonde Convolution**

$$\sum_{k=0}^r \binom{m}{k} \binom{n}{r-k} = \binom{m+n}{r}$$

14. **Upward Hockey-Stick Identity**

$$\sum_{i=r}^n \binom{i}{r} = \binom{n+1}{r+1}$$

15. **Sum of Squared Binomial Coefficients**

$$\sum_{i=0}^k \binom{k}{i}^2 = \binom{2k}{k}$$

16. **Chu-Vandermonde Special Case**

$$\sum_{i=1}^n \binom{n}{i} \binom{n-1}{i-1} = \binom{2n-1}{n-1}$$

17. **Double Subset Selection**

$$\sum_{k=q}^n \binom{n}{k} \binom{k}{q} = 2^{n-q} \binom{n}{q}$$

18. **Generalized Binomial Theorem**

$$\sum_{i=0}^n k^i \binom{n}{i} = (k+1)^n$$

19. **Half Binomial Sum**

$$\sum_{i=0}^n \binom{2n}{i} = 2^{2n-1} + \frac{1}{2} \binom{2n}{n}$$

20. **Squared Central Binomial Sum**

$$\sum_{i=0}^n \binom{2n}{i}^2 = \frac{1}{2} \left(\binom{4n}{2n} + \binom{2n}{n}^2 \right)$$